

Project Report: Temp sensor with posting data on cloud

1. Overview

- **1.1. Introduction:**

- This project implements temperature monitoring system using aht25 temperature sensor and an STM32 microcontroller.
- The system is designed for ease of use and provides real-time temperature values through mqtt.

- **1.2. Functionality Summary:**

- The system uses aht25 temp sensor to monitor the tempratue.
- An STM32 microcontroller gets the data from the sensor using I2C and converts it into a decimal value
- The converted value it then outputed from STM32 and provided to our ruggedboard using uart.
- Ruggedboard collect the data through uart and sends it to mqtt server.

2. Components and Hardware Setup

- **2.1. Hardware Components:**

- STM32 Microcontroller (STM32F446re)
- AHT25 Temp sensor
- Ruggedboard(rba5d2x)
- Power Supply
- Connecting Wires

- **2.2. Hardware Connections:**

- connections aht25 stm 32
- 1. vcc -> 5v
- 2. SDA -> PC9
- 3. gnd -> gnd
- 4. SCL -> PA8
- stm 32 to rugged board
- 1. PB10 -> mikro bus rx.

3. Software Implementation

- **3.1. Development Environment:**

- STM32CubeIDE
- HAL libraries for STM32

- Paho mqtt for mqtt operation
- Termios for uart in ruggedboard
- **3.2. Key Functions and Algorithms:**
 - **Temperature Measurement:**
 - Sends command to read status register.
 - Reads status data.
 - Sends initialization sequence.
 - Reads raw temperature and humidity data.
 - Extracts raw temperature and converts it to degrees Celsius.
 - **UART Data transfer:**
 - Send data to ruggedboard through uart.
 - **MQTT Update:**
 - Data received through uart upload to mqtt.

4. Conclusion

- **Summary of Achievements:**
 - Successfully implemented temperature measurement using aht25.
 - Achieved communication between stm32 and ruggedboard using uart.
 - Implemented uploading of the data to the cloud.